

UFUG 2106 : Discrete Mathematics

Chapter 03: The Logic of Quantified Statements

Outline

1. Predicates and Quantified Statements I
2. Predicates and Quantified Statements II
3. Statements with Multiple Quantifiers
4. Arguments with Quantified Statements

Learning Outcomes

- **Foundational Concepts**

- Identify the basic components of predicate logic: predicates, quantifiers, and domains

- **Translation and Symbolization**

- Translate natural language statements into predicate logic notation and vice versa, accurately applying quantifiers, predicates, and logical connectives.

- **Quantifiers**

- Determine the scope of quantifiers and differentiate between free and bound variables.
- Apply negation to quantified statements using De Morgan's laws for quantifiers
- Analyze statements with nested quantifiers

- **Arguments with Quantified Statements**

- Determine whether a predicate logic formula is valid (true in all interpretations) or satisfiable (true in at least one interpretation)
- Apply key equivalences and inference rules to manipulate and simplify formulas
- Construct formal proofs involving quantified statements using direct methods, contrapositive, or contradiction

Motivation

- In *Chapter 2* we learned *propositional logic* that studies statements with $\{\sim, \wedge, \vee, \rightarrow, \leftrightarrow\}$.
 - We can use propositional logic to make deductive reasoning and solve some puzzles!
- But consider the following logic problems:
 - All students in UFUG 2106 passed the exam.
 - Jack is a student of UFUG 2106.
 - Did Jack pass the exam?
- Propositional logic cannot be easily applied to the above problems.

Predicates

Definition

A **predicate** is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables. The **domain** of a predicate variable is the set of all values that may be substituted in place of the variable.

Example 3.1.1 - *Finding Truth Values of a Predicate*

Let $P(x)$ be the predicate “ $x^2 > x$ ” with domain the set $\mathbf{R}$ of all real numbers. Write $P(2)$, $P(\frac{1}{2})$, and $P(-\frac{1}{2})$, and indicate which of these statements are true and which are false.

Truth Set of a Predicate

Definition

If $P(x)$ is a predicate and x has domain D , the **truth set** of $P(x)$ is the set of all elements of D that make $P(x)$ true when they are substituted for x . The truth set of $P(x)$ is denoted

$$\{x \in D \mid P(x)\}.$$

Example 3.1.2 – *Finding the Truth Set of a Predicate*

Let $Q(n)$ be the predicate “ n is a factor of 8.” Find the truth set of $Q(n)$ if

- the domain of n is $\mathbf{Z}^+$, the set of all positive integers
- the domain of n is $\mathbf{Z}$, the set of all integers.

Quantifiers

- One sure way to change predicates into statements is to assign specific values to all their variables.
- For example, if x represents the number 35, the sentence “ x is (evenly) divisible by 5” is a true statement since $35 = 5 \cdot 7$.
 - Another way to obtain statements from predicates is to add **quantifiers**.
- **Quantifiers** are words that refer to quantities such as “some” or “all” and tell for how many elements a given predicate is true.

The Universal Quantifier: $\forall$

- The symbol $\forall$ is called the universal quantifier. Depending on the context, it is read as “for every,” “for each,” “for any,” “given any,” or “for all.”
- For example, another way to express the sentence “Every human being is mortal” or “All human beings are mortal” is to write
 - $\forall$ human beings x , x is mortal,
- which you would read as “For every human being x , x is mortal.”

The Universal Quantifier: $\forall$

- If you let H be the set of all human beings, then you can symbolize the statement more formally by writing

$$\forall x \in H, x \text{ is mortal.}$$

Definition

Let $Q(x)$ be a predicate and D the domain of x . A **universal statement** is a statement of the form “ $\forall x \in D, Q(x)$.” It is defined to be true if, and only if, $Q(x)$ is true for each individual x in D . It is defined to be false if, and only if, $Q(x)$ is false for at least one x in D . A value for x for which $Q(x)$ is false is called a **counterexample** to the universal statement.

Example 3.1.3 – *Truth and Falsity of Universal Statements*

- Let $D = \{1, 2, 3, 4, 5\}$, and consider the statement

$$\forall x \in D, x^2 \geq x.$$

- Write one way to read this statement out loud, and show that it is true.
-
- Consider the statement
- Find a counterexample $\forall x \in \mathbf{R}, x^2 \geq x$. This statement is false.

The Existential Quantifier: $\exists$

The symbol $\exists$ denotes “there exists” and is called the **existential quantifier**.

For example, the sentence “There is a student in Math 140” can be written as

$\exists$ a person p such that p is a student in Math 140,

or, more formally,

$\exists p \in P$ such that p is a student in Math 140,

where P is the set of all people.

The Existential Quantifier: $\exists$

- The domain of the predicate variable is generally indicated either between the $\exists$ symbol and the variable name or immediately following the variable name, and the words *such that* are inserted just before the predicate.

Definition

Let $Q(x)$ be a predicate and D the domain of x . An **existential statement** is a statement of the form “ $\exists x \in D$ such that $Q(x)$.” It is defined to be true if, and only if, $Q(x)$ is true for at least one x in D . It is false if, and only if, $Q(x)$ is false for all x in D .

Example 3.1.4 – *Truth and Falsity of Existential Statements*

a. Consider the statement

$$\exists m \in \mathbb{Z}^+ \text{ such that } m^2 = m.$$

Write one way to read this statement out loud, and show that it is true.

b. Let $E = \{5, 6, 7, 8\}$ and consider the statement

$$\exists m \in E \text{ such that } m^2 = m.$$

Show that this statement is false.

Formal vs. Informal Language

- It is important to be able to translate from formal to informal language when trying to make sense of mathematical concepts that are new to you.
- It is equally important to be able to translate from informal to formal language when thinking out a complicated problem.

Example 3.1.7 – Translating from Informal to Formal Language

- Rewrite each of the following statements formally. Use quantifiers and variables.
 - a. All triangles have three sides.
 - b. No dogs have wings.
 - c. Some programs are structured.

Universal Conditional Statements

- A reasonable argument can be made that the most important form of statement in mathematics is the **universal conditional statement**:

$\forall x, \text{ if } P(x) \text{ then } Q(x).$

Example 3.1.9 – Writing Universal Conditional Statements Formally

- Rewrite each of the following statements in the form

$\forall$ _____, if _____ then _____.

- a. If a real number is an integer, then it is a rational number.
- b. All bytes have eight bits.
- c. No fire trucks are green.

Equivalent Forms of Universal and Existential Statements

Observe that the two statements “ $\forall$ real number x , if x is an integer then x is rational” and “ $\forall$ integer x , x is rational” mean the same thing because the set of integers is a subset of the set of real numbers.

Both have informal translations “All integers are rational.” In fact, a statement of the form $\forall x \in U, \text{ if } P(x) \text{ then } Q(x)$

can always be rewritten in the form $\forall x \in D, Q(x)$

by narrowing U to be the subset D consisting of all values of the variable x that make $P(x)$ true.

Equivalent Forms of Universal and Existential Statements

Conversely, a statement of the form

$$\forall x \in D, Q(x)$$

can be rewritten as

$$\forall x, \text{ if } x \text{ is in } D \text{ then } Q(x).$$

Example 3.1.10 – *Equivalent Forms for Universal Statements*

Rewrite the following statement in the two forms

“ $\forall x$, if _____ then _____” and “ $\forall$ _____ x , _____”:

All squares are rectangles.

Example 3.1.11 – *Equivalent Forms for Existential Statements*

A **prime number** is an integer greater than 1 whose only positive integer factors are itself and 1.

Consider the statement “There is an integer that is both prime and even.”

Let $\text{Prime}(n)$ be “ n is prime” and $\text{Even}(n)$ be “ n is even.” Use the notation $\text{Prime}(n)$ and $\text{Even}(n)$ to rewrite this statement in the following two forms:

- a. $\exists n$ such that _____ $\wedge$ _____.
- b. $\exists$ _____ n such that _____.

Bound Variables and Scope

- For example, consider the following statements:
 - For every integer x , $x^2 \geq 0$.
 - There exists a real number x such that $x^3 = 8$.
- Statements (1) and (2) both call the variable x , but the x in Statement (1) serves a different function from the x in Statement (2).
- We say that the variable x is **bound** by the quantifier that controls it and that its **scope** begins when the quantifier introduces it and ends at the end of the quantified statement.

Implicit Quantification

Consider the statement

If a number is an integer, then it is a rational number.

As shown earlier, this statement is equivalent to a universal statement. However, it does not contain the telltale word *all* or *every* or *any* or *each*.

The only clue to indicate its universal quantification comes from the presence of the indefinite article *a*. This is an example of *implicit* universal quantification.

Implicit Quantification

Existential quantification can also be implicit. For instance, the statement “The number 24 can be written as a sum of two even integers” can be expressed formally as “ $\exists$ even integers m and n such that $24 = m + n$.”

Notation

Let $P(x)$ and $Q(x)$ be predicates and suppose the common domain of x is D .

- The notation $P(x) \Rightarrow Q(x)$ means that every element in the truth set of $P(x)$ is in the truth set of $Q(x)$, or, equivalently, $\forall x, P(x) \rightarrow Q(x)$.
- The notation $P(x) \Leftrightarrow Q(x)$ means that $P(x)$ and $Q(x)$ have identical truth sets, or, equivalently, $\forall x, P(x) \leftrightarrow Q(x)$.

Example 3.1.12 – Using $\Rightarrow$ and $\Leftrightarrow$

Let

$Q(n)$ be “ n is a factor of 8,”

$R(n)$ be “ n is a factor of 4,”

$S(n)$ be “ $n < 5$ and $n \neq 3$,”

and suppose the domain of n is $\mathbb{Z}^+$ the set of positive integers.

Use the $\Rightarrow$ and $\Leftrightarrow$ symbols to indicate true relationships among $Q(n)$, $R(n)$, and $S(n)$.

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Negations of Quantified Statements

- The general form of the negation of a universal statement follows immediately from the definitions of negation and of the truth values for universal and existential statements.

Theorem 3.2.1 Negation of a Universal Statement

The negation of a statement of the form

$$\forall x \text{ in } D, Q(x)$$

is logically equivalent to a statement of the form

$$\exists x \text{ in } D \text{ such that } \sim Q(x).$$

Symbolically,

$$\sim(\forall x \in D, Q(x)) \equiv \exists x \in D \text{ such that } \sim Q(x).$$

Negations of Quantified Statements

Thus

The negation of a universal statement (“all are”) is logically equivalent to an existential statement (“some are not” or “there is at least one that is not”).

Note that when we speak of **logical equivalence for quantified statements**, we mean that the statements always have identical truth values no matter what predicates are substituted for the predicate symbols and no matter what sets are used for the domains of the predicate variables.

Negations of Quantified Statements

Theorem 3.2.2 Negation of an Existential Statement

The negation of a statement of the form

$$\exists x \text{ in } D \text{ such that } Q(x)$$

is logically equivalent to a statement of the form

$$\forall x \text{ in } D, \sim Q(x).$$

Symbolically,

$$\sim(\exists x \in D \text{ such that } Q(x)) \equiv \forall x \in D, \sim Q(x).$$

Thus

The negation of an existential statement (“some are”) is logically equivalent to a universal statement (“none are” or “all are not”).

Example 3.2.1 – Negating Quantified Statements

Write formal negations for the following statements:

- a. $\forall$ primes p , p is odd.
- b. $\exists$ a triangle T such that the sum of the angles of T equals 200° .

Example 3.2.2 – *More Negations*

Rewrite the following statements formally. Then write formal and informal negations.

- a. No politicians are honest.
- b. The number 1,357 is not divisible by any integer between 1 and 37.

Negations of Universal Conditional Statements

- Negations of universal conditional statements are of special importance in mathematics. The form of such negations can be derived from facts that have already been established.
- By definition of the negation of a for all statement,
- But the $\sim(\forall x, P(x) \rightarrow Q(x)) \equiv \exists x \text{ such that } \sim(P(x) \rightarrow Q(x)).$ 3.2.1 is logically equivalent to an and statement. More precisely,

$$\sim(P(x) \rightarrow Q(x)) \equiv P(x) \wedge \sim Q(x). \quad \text{3.2.2}$$

Negations of Universal Conditional Statements

- Substituting (3.2.2) into (3.2.1) gives

$$\sim(\forall x, P(x) \rightarrow Q(x)) \equiv \exists x \text{ such that } (P(x) \wedge \sim Q(x)).$$

- Written somewhat less symbolically, this becomes

Negation of a Universal Conditional Statement

$$\sim(\forall x, \text{if } P(x) \text{ then } Q(x)) \equiv \exists x \text{ such that } P(x) \text{ and } \sim Q(x).$$

Example 3.2.4 – *Negating Universal Conditional Statements*

Write a formal negation for statement (a) and an informal negation for statement (b).

a. $\forall$ person p , if p is blond then p has blue eyes.

b. If a computer program has more than 100,000 lines, then it contains a bug.

The Relation among $\forall$, $\exists$, $\wedge$, and $\vee$

- The negation of a *for all* statement is a *there exists* statement, and the negation of a *there exists* statement is a *for all* statement.
- These facts are analogous to De Morgan's laws, which state that the negation of an *and* statement is an *or* statement and that the negation of an *or* statement is an *and* statement. This similarity is not accidental.
- In a sense, universal statements are generalizations of *and* statements, and existential statements are generalizations of *or* statements.

The Relation among $\forall$, $\exists$, $\wedge$, and $\vee$

If $Q(x)$ is a predicate and the domain D of x is the set $\{x_1, x_2, \dots, x_n\}$, then the statements

$$\forall x \in D, Q(x) \quad \text{and} \quad Q(x_1) \wedge Q(x_2) \wedge \dots \wedge Q(x_n)$$

are logically equivalent. For example, let $Q(x)$ be “ $x \cdot x = x$ ” and suppose $D = \{0, 1\}$. Then

$$\forall x \in D, Q(x)$$

can be rewritten as

$$\forall \text{ binary digits } x, x \cdot x = x.$$

The Relation among $\forall$, $\exists$, $\wedge$, and $\vee$

This is equivalent to

$$0 \cdot 0 = 0 \text{ and } 1 \cdot 1 = 1,$$

which can be rewritten in symbols as

$$Q(0) \wedge Q(1).$$

Similarly, if $Q(x)$ is a predicate and $D = \{x_1, x_2, \dots, x_n\}$, then the statements

are logically $\exists x \in D$ such that $Q(x)$ and $Q(x_1) \vee Q(x_2) \vee \dots \vee Q(x_n)$

The Relation among $\forall$, $\exists$, $\wedge$, and $\vee$

For example, let $Q(x)$ be “ $x + x = x$ ” and suppose $D = \{0, 1\}$. Then

$$\exists x \in D \text{ such that } Q(x)$$

can be rewritten as

$$\exists \text{ a binary digit } x \text{ such that } x + x = x.$$

This is equivalent to

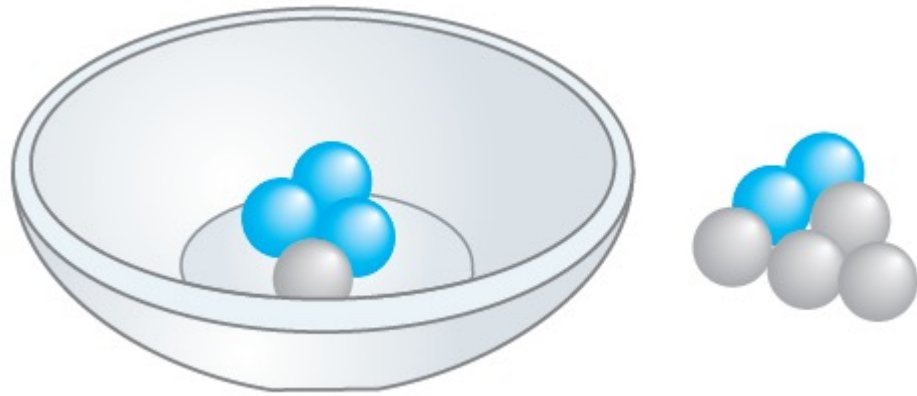
$$0 + 0 = 0 \text{ or } 1 + 1 = 1,$$

which can be rewritten in symbols as

$$Q(0) \vee Q(1).$$

Vacuous Truth of Universal Statements

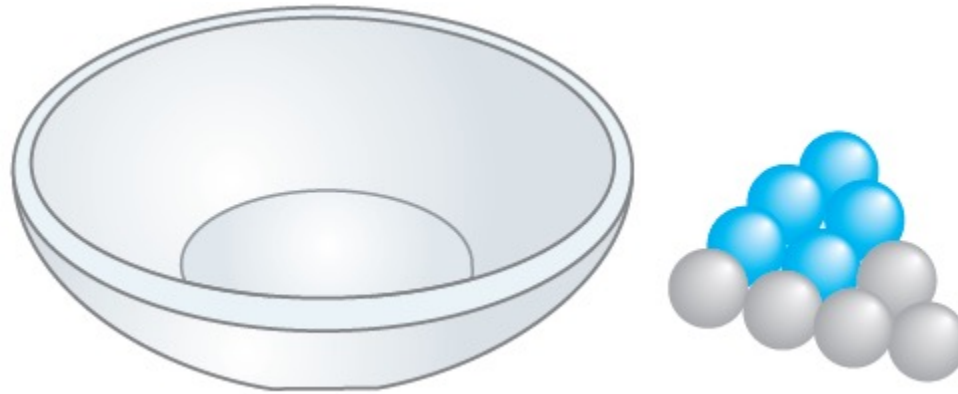
Suppose a bowl sits on a table and next to the bowl is a pile of five blue and five gray balls, any of which may be placed in the bowl. If three blue balls and one gray ball are placed in the bowl, as shown in Figure 3.2.1(a), the statement “All the balls in the bowl are blue” would be false (since one of the balls in the bowl is gray).



•Figure 3.2.1(a)

Vacuous Truth of Universal Statements

Now suppose that no balls at all are placed in the bowl, as shown in Figure 3.2.1(b).



•Figure 3.2.1(b)

Vacuous Truth of Universal Statements

Consider the statement

All the balls in the bowl are blue.

Is this statement true or false? The statement is false if, and only if, its negation is true. And its negation is

There exists a ball in the bowl that is not blue.

But the only way this negation can be true is for there actually to be a nonblue ball in the bowl. And there is not! Hence the negation is false, and so the statement is true “by default.”

Vacuous Truth of Universal Statements

In general, a statement of the form

$$\forall x \text{ in } D, \text{ if } P(x) \text{ then } Q(x)$$

is called **vacuously true** or **true by default** if, and only if, $P(x)$ is false for every x in D .

Variants of Universal Conditional Statements

- A conditional statement has a contrapositive, a converse, and an inverse. The definitions of these terms can be extended to universal conditional statements.

Definition

Consider a statement of the form $\forall x \in D$, if $P(x)$ then $Q(x)$.

1. Its **contrapositive** is the statement $\forall x \in D$, if $\sim Q(x)$ then $\sim P(x)$.
2. Its **converse** is the statement $\forall x \in D$, if $Q(x)$ then $P(x)$.
3. Its **inverse** is the statement $\forall x \in D$, if $\sim P(x)$ then $\sim Q(x)$.

Example 3.2.5 – Contrapositive, Converse, and Inverse of a Universal Conditional Statement

Write a formal and an informal contrapositive, converse, and inverse for the following statement:

If a real number is greater than 2, then its square is greater than 4.

Necessary and Sufficient Conditions, Only If

- The definitions of *necessary*, *sufficient*, and *only if* can also be extended to apply to universal conditional statements.

Definition

- “ $\forall x, r(x)$ is a **sufficient condition** for $s(x)$ ” means “ $\forall x$, if $r(x)$ then $s(x)$.”
- “ $\forall x, r(x)$ is a **necessary condition** for $s(x)$ ” means “ $\forall x$, if $\sim r(x)$ then $\sim s(x)$ ” or, equivalently, “ $\forall x$, if $s(x)$ then $r(x)$.”
- “ $\forall x, r(x)$ **only if** $s(x)$ ” means “ $\forall x$, if $\sim s(x)$ then $\sim r(x)$ ” or, equivalently, “ $\forall x$, if $r(x)$ then $s(x)$.”

Example 3.2.6 – Necessary and Sufficient Conditions

Rewrite each of the following as a universal conditional statement, quantified either explicitly or implicitly. Do not use the word *necessary* or *sufficient*.

- a. Squareness is a sufficient condition for rectangularity.
- b. Being at least 35 years old is a necessary condition for being president of the United States.

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Statements with Multiple Quantifiers

When a statement contains more than one kind of quantifier, we imagine the actions suggested by the quantifiers as being performed in the order in which the quantifiers occur. For instance, consider a statement of the form

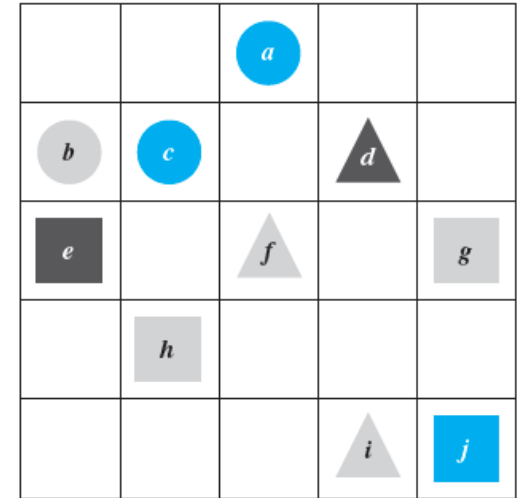
$\forall x$ in set D , $\exists y$ in set E such that x and y satisfy property $P(x, y)$.

- To show that such a statement is true, you must be able to meet the following challenge:
 - Imagine that someone is allowed to choose any element whatsoever from the set D , and imagine that the person gives you that element. Call it x .
 - The challenge for you is to find an element y in E so that the person's x and your y , taken together, satisfy property $P(x, y)$.
- ***Because you do not have to specify the y until after the other person has specified the x , you are allowed to find a different value of y for each different x you are given.***

Example 3.3.1 – *Truth of a $\forall\exists$ Statement in a Tarski World*

- Consider the Tarski world shown in Figure 3.3.1. Show that the following statement is true in this world:

For every triangle x , there is a square y such that x and y have the same color.

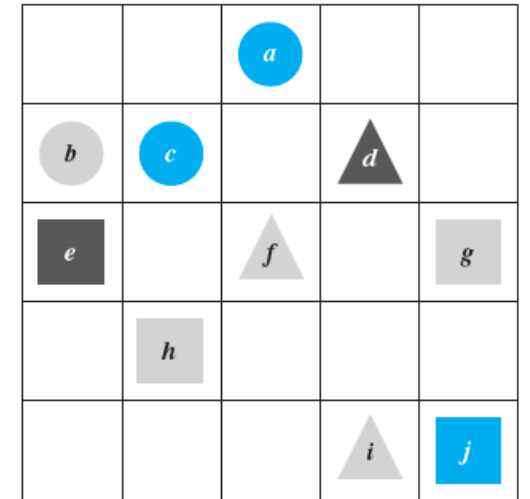


•Figure 3.3.1

Example 3.3.2 – Truth of a $\exists\forall$ Statement in a Tarski World

Consider the Tarski world shown in Figure 3.3.1. Show that the following statement is true:

There is a triangle x such that for every circle y , x is to the right of y .



•Figure 3.3.1

Statements with Multiple Quantifiers

Interpreting Statements with Two Different Quantifiers

If you want to establish the truth of a statement of the form

$$\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } P(x, y)$$

your challenge is to allow someone else to pick whatever element x in D they wish and then you must find an element y in E that “works” for that particular x .

If you want to establish the truth of a statement of the form

$$\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, P(x, y)$$

your job is to find one particular x in D that will “work” no matter what y in E anyone might choose to challenge you with.

Example 3.3.3 – *Interpreting Statements with More Than One Quantifier*

A college cafeteria line has four stations: salads, main courses, desserts, and beverages. The salad station offers a choice of green salad or fruit salad; the main course station offers spaghetti or fish; the dessert station offers pie or cake; and the beverage station offers milk, soda, or coffee. Three students, Uta, Tim, and Yuen, go through the line and make the following choices:

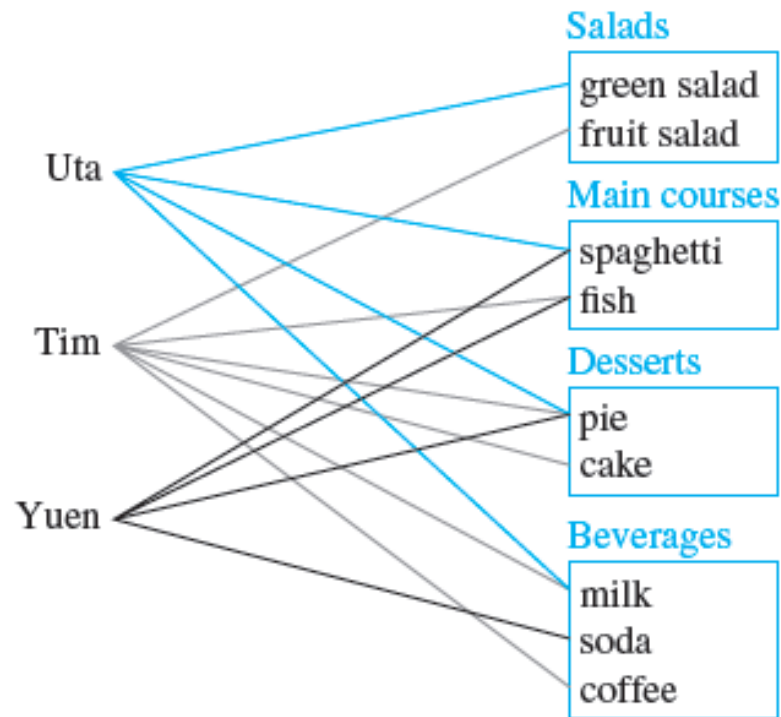
Uta: green salad, spaghetti, pie, milk

Tim: fruit salad, fish, pie, cake, milk, coffee

Yuen: spaghetti, fish, pie, soda

Example 3.3.3 – *Interpreting Statements with More Than One Quantifier* continued

These choices are illustrated in Figure 3.3.2.



•Figure 3.3.2

Example 3.3.3 – *Interpreting Statements with More Than One Quantifier* continued

Write each of following statements informally and find its truth value.

- a. $\exists$ an item I such that $\forall$ student S , S chose I .
- b. $\exists$ a student S such that $\forall$ item I , S chose I .
- c. $\exists$ a student S such that $\forall$ station Z , $\exists$ an item I in Z such that S chose I .
- d. $\forall$ student S and $\forall$ station Z , $\exists$ an item I in Z such that S chose I .

Translating from Informal to Formal Language

Most problems are stated in informal language, but solving them often requires translating them into more formal terms.

Example 3.3.4 – Translating Statements with Multiple Quantifiers from Informal to Formal Language

The **reciprocal** of a real number a is a real number b such that $ab = 1$. The following two statements are true. Rewrite them formally using quantifiers and variables.

- a. Every nonzero real number has a reciprocal.
- b. There is a real number with no reciprocal.

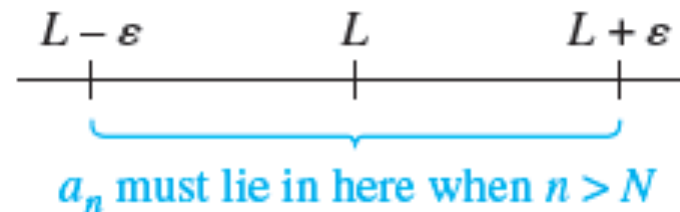
Example 3.3.7 – *The Definition of Limit of a Sequence*

The definition of limit of a sequence, studied in calculus, uses both quantifiers $\forall$ and $\exists$ and also if-then. We say that the limit of the sequence a_n as n goes to infinity equals L and write

$$\lim_{n \rightarrow \infty} a_n = L$$

if, and only if, the values of a_n become *arbitrarily* close to L as n gets larger and larger without bound.

More precisely, this means that given any positive number ε , we can find an integer N such that whenever n is larger than N , the number a_n sits between $L - \varepsilon$ and $L + \varepsilon$ on the number line.



Example 3.3.7 – *The Definition of Limit of a Sequence*

continued

- Symbolically:

- $\forall \varepsilon > 0, \exists$ an integer N such that $\forall$ integer n ,
 - if $n > N$ then $L - \varepsilon < a_n < L + \varepsilon$.
- Considering the logical complexity of this definition, it is no wonder that many students find it hard to understand.

Negations of Statements with More Than One Quantifier

You can use the same rules to negate statements with several quantifiers that you used to negate simpler quantified statements. We know that

and

$$\sim(\forall x \text{ in } D, P(x)) \equiv \exists x \text{ in } D \text{ such that } \sim P(x).$$

Thus

$$\sim(\exists x \text{ in } D \text{ such that } P(x)) \equiv \forall x \text{ in } D, \sim P(x).$$

$$\begin{aligned} \sim(\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } P(x, y)) &\equiv \exists x \text{ in } D \text{ such that } \sim(\exists y \text{ in } E \text{ such that } P(x, y)) \\ &\equiv \exists x \text{ in } D \text{ such that } \forall y \text{ in } E, \sim P(x, y) \end{aligned}$$

Negations of Statements with More Than One Quantifier

Similarly,

$$\begin{aligned}\sim(\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, P(x, y)) &\equiv \forall x \text{ in } D, \sim(\forall y \text{ in } E, P(x, y)) \\ &\equiv \forall x \text{ in } D, \exists y \text{ in } E \text{ such that } \sim P(x, y)\end{aligned}$$

These facts are summarized as follows:

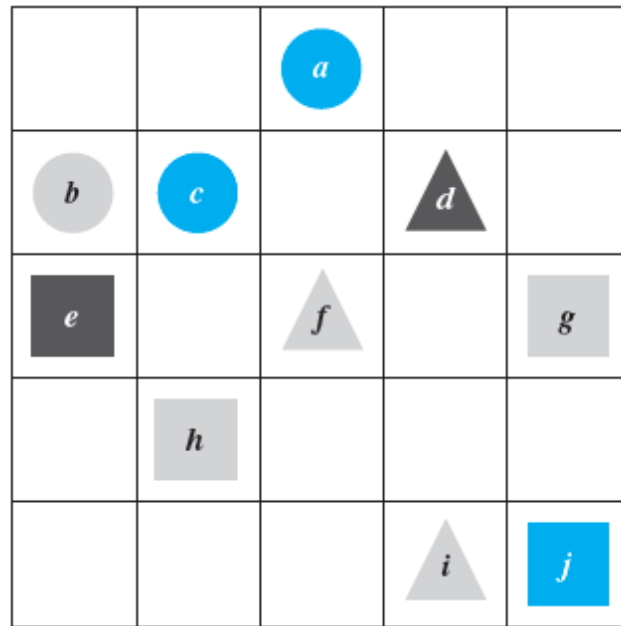
Negations of Statements with Two Different Quantifiers

$$\sim(\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } P(x, y)) \equiv \exists x \text{ in } D \text{ such that } \forall y \text{ in } E, \sim P(x, y)$$

$$\sim(\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, P(x, y)) \equiv \forall x \text{ in } D, \exists y \text{ in } E \text{ such that } \sim P(x, y)$$

Example 3.3.8 – *Negating Statements in a Tarski World*

Consider the Tarski world shown in Figure 3.3.1.



•Figure 3.3.1

Example 3.3.8 – *Negating Statements in a Tarski World* continued

Write a negation for each of the following statements, and determine which is true, the given statement or its negation.

- a. For every square x , there is a circle y such that x and y have the same color.
- b. There is a triangle x such that for every square y , x is to the right of y .

Order of Quantifiers

Consider the following two statements:

$\forall$ person x , $\exists$ a person y such that x loves y .

$\exists$ a person y such that $\forall$ person x , x loves y .

Note that except for the order of the quantifiers, these statements are identical.

However, the first means that given any person, it is possible to find someone whom that person loves, whereas the second means that there is one amazing individual who is loved by all people.

Order of Quantifiers

The two sentences illustrate an extremely important property about statements with two different quantifiers.

In a statement containing both $\forall$ and $\exists$, changing the order of the quantifiers can significantly change the meaning of the statement.

Interestingly, however, if one quantifier immediately follows another quantifier *of the same type*, then the order of the quantifiers does not affect the meaning.

Order of Quantifiers

Consider the commutative property of addition of real numbers, for example:

$\forall$ real number x and $\forall$ real number y , $x + y = y + x$.

This means the same as

$\forall$ real number y and $\forall$ real number x , $x + y = y + x$.

Thus the property can be expressed a little less formally as

$\forall$ real numbers x and y , $x + y = y + x$.

Formal Logical Notation

In some areas of computer science, logical statements are expressed in purely symbolic notation.

The notation involves using predicates to describe all properties of variables and omitting the words *such that* in existential statements. The formalism also depends on the following facts:

“ $\forall x \text{ in } D, P(x)$ ” can be written as “ $\forall x (x \text{ in } D \rightarrow P(x))$,”

and

“ $\exists x \text{ in } D \text{ such that } P(x)$ ” can be written as “ $\exists x (x \text{ in } D \wedge P(x))$.”

Formal Logical Notation

The disadvantage of the fully formal notation is that because it is complex and somewhat remote from intuitive understanding, when we use it, we may make errors that go unrecognized.

The advantage, however, is that operations, such as taking negations, can be made completely mechanical and programmed on a computer.

Taken together, the symbols for quantifiers, variables, predicates, and logical connectives make up what is known as the **language of first-order logic**.

Summary of Two Quantifiers

TABLE 1 Quantifications of Two Variables.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Outline

1. Predicates and Quantified Statements I
2. Predicates and Quantified Statements II
3. Statements with Multiple Quantifiers
4. Arguments with Quantified Statements

Arguments with Quantified Statements

The rule of *universal instantiation* (in-stan-she-AY-shun) says the following:

Universal Instantiation

If a property is true of *everything* in a set, then it is true of *any particular* thing in the set.

Use of the words *universal instantiation* indicates that the truth of a property in a particular case follows as a special instance of its more general or universal truth.

Arguments with Quantified Statements

The validity of this argument form follows immediately from the definition of truth values for a universal statement. One of the most famous examples of universal instantiation is the following:

All men are mortal.

Socrates is a man.

∴ Socrates is mortal.

Arguments with Quantified Statements

As an example of universal instantiation, suppose you are doing a problem that requires you to simplify

$$r^{k+1} \cdot r,$$

where r is a particular real number and k is a particular integer. You know from your study of algebra that the following universal statements are true:

1. For every real number x and for all integers m and n , $x^m \cdot x^n = x^{m+n}$.
2. For every real number x , $x^1 = x$.

Arguments with Quantified Statements

So you proceed as follows:

$$\begin{aligned} r^{k+1} \cdot r &= r^{k+1} \cdot r^1 && \text{Step 1} \\ &= r^{(k+1)+1} && \text{Step 2} \\ &= r^{k+2} && \text{by basic algebra.} \end{aligned}$$

Here is the reasoning behind steps 1 and 2.

Step 1: For every real number x , $x^1 = x$. universal truth

r is a particular real number. particular instance

$\therefore r^1 = r$. conclusion

Arguments with Quantified Statements

Step 2: For every real number x and for all integers m and n , $x^m \cdot x^n = x^{m+n}$. universal truth

$$n, \quad x^m \cdot x^n = x^{m+n}.$$

r is a particular real number and $k + 1$ and 1 are particular integers. particular instance

$$\therefore r^{k+1} \cdot r^1 = r^{(k+1)+1}.$$

conclusion

Both arguments are examples of universal instantiation.

Universal Modus Ponens

The rule of universal instantiation can be combined with modus ponens to obtain the valid form of argument called *universal modus ponens*.

Universal Modus Ponens

Formal Version

$\forall x$, if $P(x)$ then $Q(x)$.
 $P(a)$ for a particular a .
 $\therefore Q(a)$.

Informal Version

If x makes $P(x)$ true, then x makes $Q(x)$ true.
 a makes $P(x)$ true.
 $\therefore a$ makes $Q(x)$ true.

Example 3.4.2 – *Drawing Conclusions Using Universal Modus Ponens*

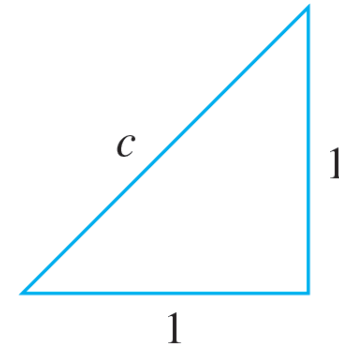
Write the conclusion that can be inferred using universal modus ponens.

If T is any right triangle with hypotenuse c and legs a and b , then $c^2 = a^2 + b^2$.

The triangle shown at the right is a right triangle with both legs equal to 1 and hypotenuse c .

∴ _____

Pythagorean Theorem



Use of Universal Modus Ponens in a Proof

Here is a proof that **the sum of any two even integers is even**.

It makes use of the definition of even integer, namely, that an integer is *even* if, and only if, it equals twice some integer.

(Or, more formally: $\forall$ integers x , x is even if, and only if, $\exists$ an integer—say, k —such that $x = 2k$.) Suppose m and n are particular but arbitrarily chosen even integers. Then $m = 2r$ for some integer r ,⁽¹⁾ and $n = 2s$ for some integer s .⁽²⁾ Hence

$$\begin{aligned} m + n &= 2r + 2s && \text{by substitution} \\ &= 2(r + s) && \text{by factoring out the 2.} \end{aligned}$$

Use of Universal Modus Ponens in a Proof

Now $r + s$ is an integer,⁽⁴⁾ and so $2(r + s)$ is even.⁽⁵⁾ Thus $m + n$ is even.

The following expansion of the proof shows how each of the numbered steps is justified by arguments that are valid by universal modus ponens.

- (1) If an integer is even, then it equals twice some integer.
 m is a particular even integer.
 $\therefore m$ equals twice some integer, say, r .

Use of Universal Modus Ponens in a Proof

(2) If an integer is even, then it equals twice some integer.

n is a particular even integer.

$\therefore n$ equals twice some integer, say, s .

(3) If a quantity is an integer, then it is a real number.

r and s are particular integers.

$\therefore r$ and s are real numbers.

For all a , b , and c , if a , b , and c are real numbers, then $ab + ac = a(b + c)$.

2 , r , and s are particular real numbers.

$\therefore 2r + 2s = 2(r + s)$.

Use of Universal Modus Ponens in a Proof

- (4) For all u and v , if u and v are integers, then $u + v$ is an integer.
 r and s are two particular integers.
 $\therefore r + s$ is an integer.
- (5) If a number equals twice some integer, then that number is even.
 $2(r + s)$ equals twice the integer $r + s$.
 $\therefore 2(r + s)$ is even.

Universal Modus Tollens

Another crucially important rule of inference is *universal modus tollens*. Its validity results from combining universal instantiation with modus tollens. **Universal modus tollens is the heart of proof of contradiction**, which is one of the most important methods of mathematical argument.

Universal Modus Tollens

Formal Version

$\forall x, \text{ if } P(x) \text{ then } Q(x).$
 $\sim Q(a), \text{ for a particular } a.$
 $\therefore \sim P(a).$

Informal Version

If x makes $P(x)$ true, then x makes $Q(x)$ true.
 a does not make $Q(x)$ true.
 $\therefore a$ does not make $P(x)$ true.

Example 3.4.3 – *Recognizing the Form of Universal Modus Tollens*

Rewrite the following argument using quantifiers, variables, and predicate symbols. Write the major premise in conditional form. Is this argument valid? Why?

All human beings are mortal.

Zeus is not mortal.

∴ Zeus is not human.

Converse Error and Inverse Error

Converse Error (Quantified Form)

Formal Version

$\forall x, \text{ if } P(x) \text{ then } Q(x).$
 $Q(a) \text{ for a particular } a.$
 $\therefore P(a).$ ← invalid
conclusion

Informal Version

If x makes $P(x)$ true, then x makes $Q(x)$ true.
 a makes $Q(x)$ true.
 $\therefore a$ makes $P(x)$ true. ← invalid
conclusion

Inverse Error (Quantified Form)

Formal Version

$\forall x, \text{ if } P(x) \text{ then } Q(x).$
 $\sim P(a), \text{ for a particular } a.$
 $\therefore \sim Q(a).$ ← invalid
conclusion

Informal Version

If x makes $P(x)$ true, then x makes $Q(x)$ true.
 a does not make $P(x)$ true.
 $\therefore a$ does not make $Q(x)$ true. ← invalid
conclusion

Creating Additional Forms of Argument

Universal modus ponens and modus tollens were obtained by combining universal instantiation with modus ponens and modus tollens. In the same way, additional forms of arguments involving universally quantified statements can be obtained by combining universal instantiation with other of the valid argument forms.

Universal Transitivity

Formal Version

$\forall x P(x) \rightarrow Q(x).$
 $\forall x Q(x) \rightarrow R(x).$
 $\therefore \forall x P(x) \rightarrow R(x).$

Informal Version

Any x that makes $P(x)$ true makes $Q(x)$ true.
Any x that makes $Q(x)$ true makes $R(x)$ true.
 $\therefore$ Any x that makes $P(x)$ true makes $R(x)$ true.

Rules of Inference for Quantified Statements

TABLE 2 Rules of Inference for Quantified Statements.

<i>Rule of Inference</i>	<i>Name</i>
$\frac{\forall xP(x)}{\therefore P(c)}$	Universal instantiation
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall xP(x)}$	Universal generalization
$\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists xP(x)}$	Existential generalization

Example of Argument with Predicates

Let $C(x)$ be “ x is in this class”, $B(x)$ be “ x has read the book”, $P(x)$ be “ x passed the first exam”.

Deduce the conclusion from the following given premises:

$$\textcircled{1} \exists x(C(x) \wedge \sim B(x))$$

$$\textcircled{2} \forall x(C(x) \rightarrow P(x))$$

Conclusion: $\exists x(P(x) \wedge \sim B(x))$

Solution:

Step

Reason

$$1. \exists x(C(x) \wedge \neg B(x))$$

Premise

$$2. C(a) \wedge \neg B(a)$$

Existential instantiation from (1)

$$3. C(a)$$

Simplification from (2)

$$4. \forall x(C(x) \rightarrow P(x))$$

Premise

$$5. C(a) \rightarrow P(a)$$

Universal instantiation from (4)

$$6. P(a)$$

Modus ponens from (3) and (5)

$$7. \neg B(a)$$

Simplification from (2)

$$8. P(a) \wedge \neg B(a)$$

Conjunction from (6) and (7)

$$9. \exists x(P(x) \wedge \neg B(x))$$

Existential generalization from (8)

Example of Argument with Predicates

Deduce the conclusion from the following given premises:

- ① If anyone likes walking, he doesn't like car.
- ② Anyone likes car or bicycle.
- ③ Someone doesn't like bicycle.

Conclusion: someone doesn't like walking.

Let $W(x)$ denote “ x likes walking”, $C(x)$ denote “ x likes car”, $B(x)$ denote “ x likes bicycle”.

The premises are: (1) $\forall x(W(x) \rightarrow \sim C(x))$; (2) $\forall x(C(x) \vee B(x))$; (3) $\exists x \sim B(x)$

The conclusion is: $\exists x \sim W(x)$